

Resource-Constrained Competition: A Mathematical Bridge Between Economics and Ecology

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Abstract

We demonstrate that market equilibria in economic theory and ecological equilibria in population dynamics are mathematically isomorphic solutions to resource allocation problems. By formalizing both frameworks as dynamical systems with resource constraints, we show that monopoly, duopoly, oligopoly, and perfect competition correspond precisely to single-species, two-species, multi-species, and intense competition scenarios in ecology. This structural equivalence reveals a general principle: systems where agents compete for scarce resources converge to predictable configurations regardless of whether allocation occurs through conscious optimization or evolutionary selection.

1. Introduction

Competition for scarce resources is fundamental to both economic and ecological systems. In markets, firms compete for consumer demand; in ecosystems, species compete for nutrients, space, and energy. While these domains appear distinct, economics deals with rational agents maximizing profits, ecology with populations subject to natural selection, both address the same core problem: how do multiple competitors allocate a limited resource among themselves?

Game theory provides the mathematical framework for analyzing strategic interactions in economics, yielding equilibrium concepts such as Nash equilibrium. Population ecology employs dynamical systems, particularly Lotka-Volterra equations, to describe competitive interactions. Despite different historical origins and methodologies, we demonstrate that these frameworks produce identical mathematical structures.

Our contribution is threefold: (1) we establish precise formal correspondences between economic and ecological variables, (2) we show that standard market structures (monopoly through perfect competition) map directly onto ecological competition scenarios, and (3) we propose a unified dynamical system that encompasses both domains as special cases. Crucially, we demonstrate that the static equilibrium conditions from game theory can be transformed into the dynamical equations of ecology, and vice versa, revealing that these are alternative representations of the same underlying structure.

2. Market Equilibria as Optimization Problems

Consider first the economic framework where firms choose quantities to maximize profits. A monopolist faces inverse demand function $P(q)$ and cost function $C(q)$, solving $\max_q \pi = q \cdot P(q) - C(q)$. The first-order condition yields equilibrium quantity q^* where marginal revenue equals

marginal cost. This represents the simplest case: a single agent optimally exploiting a resource without competition.

In a Cournot duopoly, two firms simultaneously choose quantities q_1, q_2 . Firm i 's profit is $\pi_i = q_i \cdot P(q_1 + q_2) - C_i(q_i)$. Nash equilibrium requires both firms to simultaneously optimize: $\frac{\partial \pi_1}{\partial q_1} = 0$ and $\frac{\partial \pi_2}{\partial q_2} = 0$. These conditions define reaction functions whose intersection determines equilibrium (derivation in Appendix A).

The oligopoly with n firms generalizes to equilibrium conditions $\frac{\partial \pi_i}{\partial q_i} = 0$ for all $i \in \{1, \dots, n\}$, a system of n simultaneous equations. As $n \rightarrow \infty$ with free entry, the system approaches perfect competition where price equals marginal cost and profits vanish. Only the most efficient firms survive.

These equilibrium conditions are typically derived as static optimization problems. However, we can equivalently represent them as dynamical systems where firms adjust quantities based on profit gradients: $\frac{dq_i}{dt} = \lambda q_i \frac{\partial \pi_i}{\partial q_i}$ where $\lambda > 0$ is an adjustment speed parameter. This gradient ascent process converges to Nash equilibrium where $\frac{\partial \pi_i}{\partial q_i} = 0$. The adjustment speed λ determines convergence rate toward equilibrium and plays an analogous role to the intrinsic growth rate r in ecological dynamics (discussed in Section 3). Local stability of this dynamical system can be verified through Lyapunov analysis: the profit function serves as a potential, and its gradient defines the flow. This dynamical formulation is essential for connecting to ecological models.

3. Ecological Competition as Dynamical Systems

Population ecology naturally employs dynamical systems. A single species with population N , intrinsic growth rate r , and carrying capacity K evolves according to the logistic equation $\frac{dN}{dt} = rN(1 - \frac{N}{K})$. Growth slows as resources become scarce, reaching equilibrium at $N^* = K$ where birth equals death rate.

When two species compete, we use Lotka-Volterra equations:

$$\frac{dN_1}{dt} = r_1 N_1 (1 - \frac{N_1 + \alpha_{12} N_2}{K_1}), \quad \frac{dN_2}{dt} = r_2 N_2 (1 - \frac{N_2 + \alpha_{21} N_1}{K_2})$$

The competition coefficient α_{ij} measures how much species j depletes resources available to species i . The term $N_1 + \alpha_{12} N_2$ represents effective resource consumption from species 1's perspective. Equilibrium occurs where both growth rates vanish, at the intersection of zero-growth isoclines.

For n species: $\frac{dN_i}{dt} = r_i N_i (1 - \frac{N_i + \sum_{j \neq i} \alpha_{ij} N_j}{K_i})$. Coexistence requires moderate competition; intense competition leads to exclusion where only the most efficient species survives ($N_j \rightarrow 0$ for less efficient competitors).

4. The Structural Correspondence

The fundamental correspondence emerges from recognizing market demand as the economic analog of ecological carrying capacity. In ecology, K represents total resource available, nutrients, space, or energy. Species compete by consuming shares of this resource. In economics, market demand represents the total "resource" available to firms. Firms compete by producing output, and total quantity $Q = \sum q_i$ cannot be absorbed without price declining.

This yields natural variable mappings: species $i \leftrightarrow$ firm i , population $N_i \leftrightarrow$ output q_i , carrying capacity $K \leftrightarrow$ market size, growth rate $r_i \leftrightarrow$ production efficiency (inversely related to marginal cost), competition coefficient $\alpha_{ij} \leftrightarrow$ strategic interaction parameter. The resource limitation, more organisms mean less per capita, parallels the demand constraint where more output means lower price per unit.

The core mechanism is identical. In ecology, increasing $\sum N_i$ depletes resources, reducing per-capita growth until $dN_i/dt = 0$. In economics, increasing $Q = \sum q_i$ lowers price $P(Q)$, reducing per-unit profit until $\frac{\partial \pi_i}{\partial q_i} = 0$. Both feature negative feedback through resource depletion or price erosion.

5. From Game Equilibria to Dynamical Systems: The Mathematical Equivalence

The connection between economic and ecological models becomes explicit when we show that game-theoretic equilibrium conditions transform into dynamical equations. Consider a Cournot duopoly with linear inverse demand $P(Q) = a - bQ$ and constant marginal costs c_i . Firm i 's profit is $\pi_i = q_i(a - bQ - c_i)$ where $Q = q_1 + q_2$. The Nash equilibrium condition $\frac{\partial \pi_i}{\partial q_i} = 0$ gives:

$$a - b(q_1 + q_2) - bq_i - c_i = 0$$

For the two firms, this yields the system:

$$q_1 + 2q_2 = \frac{a - c_1}{b}, \quad q_2 + 2q_1 = \frac{a - c_2}{b}$$

Now consider Lotka-Volterra with equal carrying capacities $K_1 = K_2 = K$. At interior equilibrium ($N_i > 0$), $\frac{dN_1}{dt} = 0$ implies:

$$N_1 + \alpha_{12}N_2 = K, \quad N_2 + \alpha_{21}N_1 = K$$

These are linear systems with identical structure. To establish formal equivalence, we require parameter mappings that preserve this structure. Under the identification $N_i \leftrightarrow q_i$ and assuming symmetric competition ($\alpha_{12} = \alpha_{21} = \alpha$), comparing coefficients gives:

$$K = \frac{a - c_i}{b(1 + \alpha)}, \quad \alpha = 1$$

The key condition is linearity: the linear inverse demand $P(Q) = a - bQ$ produces a competitive effect proportional to total output, exactly as the linear competitive term $\alpha_{12}N_2$ in Lotka-Volterra produces an effect proportional to competitor population. Under this assumption, the Cournot equilibrium conditions become:

$$q_1 + \alpha q_2 = K, \quad q_2 + \alpha q_1 = K$$

which is identical to the Lotka-Volterra equilibrium system. The identification $K = \frac{a-c_i}{b(1+\alpha)}$ ensures quantitative correspondence: the carrying capacity scales with market size (measured by demand intercept a) minus cost disadvantage, normalized by price sensitivity b and competition intensity. This formula reveals how economic parameters map to ecological constraints: a larger market (a high) or lower costs (c_i low) increase carrying capacity, while stronger competition (α high) or higher price sensitivity (b high) reduce it. The carrying capacity thus represents the sustainable scale of operation given market and competitive conditions.

This equivalence holds locally around equilibrium under linearity assumptions. Non-linear demand $P(Q)$ or cost functions $C(q)$ would require Taylor expansion around equilibrium, where the linear approximation recovers the Lotka-Volterra structure with appropriately defined effective parameters. The isomorphism is therefore structural and local, it characterizes the system's behavior in a neighborhood of equilibrium rather than globally. For strongly non-linear systems, the correspondence applies to the linearized dynamics, with higher-order terms introducing deviations that may be analyzed through perturbation methods.

The transformation works bidirectionally. Given Nash equilibrium conditions $\frac{\partial \pi_i}{\partial q_i} = 0$, we construct dynamics $\frac{dq_i}{dt} = q_i \frac{\partial \pi_i}{\partial q_i}$ whose fixed points are Nash equilibria. Conversely, given ecological equilibrium $\frac{dN_i}{dt} = 0$, we construct a payoff function such that maximizing cumulative fitness subject to resource constraints yields the ecological equilibrium as an optimum. The distinction between "static game equilibrium" and "dynamical ecological equilibrium" is presentational: Nash conditions are steady states of adjustment dynamics; ecological equilibria solve implicit optimization problems.

Theorem (Mathematical Isomorphism): Under linear inverse demand $P(Q) = a - bQ$ and constant marginal costs c_i , the Nash equilibrium conditions of the n -firm Cournot oligopoly are isomorphic to the steady-state conditions of the n -species Lotka-Volterra competition system with symmetric competition coefficients and equal carrying capacities. Specifically, there exists a bijective parameter map $\Phi: (q^*, a, b, c) \leftrightarrow (N^*, r, K, A)$ such that the equilibrium conditions are identical under this map.

This theorem formalizes the equivalence: both systems solve the same mathematical problem of allocating a limited resource among competing agents. The proof follows from the coefficient comparison performed above and is constructive, given parameters in one domain, we can explicitly compute corresponding parameters in the other.

Both optimization frameworks derive from a potential function. In economics, firms ascend a profit landscape; in ecology, species ascend a fitness landscape. The gradient of this potential, $\nabla_q \pi$ or $\nabla_N W$ (where W is the fitness potential), is formally identical to the growth/adjustment term in the

respective dynamical systems. This shared variational structure underlies the isomorphism: systems with the same potential topology must have the same equilibrium structure.

6. Equilibrium Correspondences Across Market Structures

With mathematical equivalence established, we systematically map market structures to ecological scenarios. Monopoly corresponds to single-species logistic growth: the monopolist optimally exploits market demand as a single species grows to carrying capacity $N^* = K$. Both represent optimization subject only to resource limits, with no strategic interaction.

Cournot duopoly maps to two-species Lotka-Volterra competition. Both are two-dimensional systems with isoclines determining equilibrium. Reaction functions $R_i(q_j)$ in Cournot, each firm's optimal response to competitor's quantity, are economic isoclines. Zero-growth isoclines $\frac{dN_i}{dt} = 0$ in Lotka-Volterra represent population balance given competitor's size. Equilibrium is the intersection in both cases. Stability conditions are identical. Cournot equilibrium is stable when each firm's self-limitation exceeds cross-effects from competitors. Ecologically, equilibrium is stable when intraspecific competition ($\frac{1}{K_i}$) exceeds interspecific competition (α_{ij}). Both require diagonal dominance in the interaction matrix, a standard stability criterion.

Oligopoly with n firms corresponds to n -species competition. Coexistence requires the system is not "too competitive": economically, market must be large relative to costs; ecologically, competition coefficients must be small. When these conditions fail, some competitors exit. The mathematics determining survivors is identical: solve equilibrium with various competitors set to zero, check stability and feasibility.

Perfect competition ($n \rightarrow \infty$, free entry) maps to competitive exclusion. Economic profits vanish, price equals marginal cost, only lowest-cost firms survive. Ecologically, when many species compete intensely (large α_{ij}), only the most efficient (largest r_i/K_i ratio) survives. Both predict elimination of less efficient competitors and zero surplus at equilibrium.

7. Selection, Optimization, and Convergence Mechanisms

While firms appear to optimize consciously and species evolve blindly, convergence mechanisms are formally similar. Firms adjust via imitation, profit signals, and trial-and-error, behavioral dynamics resembling evolution. Market competition eliminates unprofitable strategies as selection eliminates unfit organisms. Conversely, natural selection implements optimization: heritable variation plus differential reproduction converges to fitness-maximizing strategies.

Evolutionary game theory formalizes this. The replicator equation $\frac{dx_i}{dt} = x_i[f_i(x) - \bar{f}(x)]$ describes strategy frequency evolution, where f_i is fitness and $\bar{f}$ is average fitness. Stable fixed points are Evolutionarily Stable Strategies (ESS), which satisfy Nash equilibrium conditions mathematically. Both systems converge through iterative adjustment: quantity changes following profit gradients

(economics) or population changes following fitness differentials (ecology). The distinction is mechanism and timescale, not mathematical structure.

Predictive implication: If the parallelism holds, variations in market concentration should follow stability conditions observed in competitive ecological networks. Specifically, markets should support n firms only when $\det(I - A) > 0$ where A is the normalized interaction matrix, the same condition governing species coexistence.

Empirical measurement: This condition can be tested in real markets by estimating the interaction matrix from observed data. For firm i , the competition coefficient α_{ij} can be approximated by the cross-elasticity of firm i 's profit with respect to firm j 's output, normalized by own-effects. Specifically, $\alpha_{ij} \approx -\frac{\partial^2 \pi_i / \partial q_i \partial q_j}{\partial^2 \pi_i / \partial q_i^2}$. Econometric estimation of market share dynamics and price responses allows construction of A , enabling prediction of which market structures are sustainable. Industries with $\det(I - A) < 0$ should exhibit consolidation toward fewer firms, while those with $\det(I - A) > 0$ can sustain multiple competitors, testable predictions linking market structure to competitive intensity.

8. A Unified Framework

Both economic and ecological equilibria are special cases of a general dynamical system describing competition for limited resources. Consider n competitors with states $x_i \geq 0$. The unified system is:

$$\frac{dx_i}{dt} = x_i \cdot f_i(x; r, A)$$

where $x = (x_1, \dots, x_n)$, $r = (r_1, \dots, r_n)$ are efficiency parameters, and $A = [\alpha_{ij}]$ is the interaction matrix. The function f_i is:

$$f_i(x; r, A) = r_i \left(1 - \frac{x_i + \sum_{j \neq i} \alpha_{ij} x_j}{K} \right)$$

In ecology: $x_i = N_i$ (population), r_i is intrinsic growth rate, K is carrying capacity, α_{ij} is interspecific competition, dynamics on generational timescales.

In economics: $x_i = q_i$ (output), $r_i \propto 1/c_i$ (inverse marginal cost), $K \propto$ demand, α_{ij} captures strategic effects, dynamics on market adjustment timescales.

Equilibrium occurs when $f_i(x^*) = 0$ for all i : either $x_i^* + \sum_{j \neq i} \alpha_{ij} x_j^* = K$ (interior) or $x_i^* = 0$ (exclusion). Stability follows from the Jacobian $J_{ij} = \frac{\partial}{\partial x_j} [x_i f_i(x)]|_{x^*}$. Equilibrium is stable if all eigenvalues have negative real parts, identical across domains.

Dynamics under exogenous shocks: Consider an increase in K (expanded carrying capacity or market demand). The equilibrium shifts along predictable trajectories determined by $\frac{dx^*}{dK} = -J^{-1} \frac{\partial f}{\partial K}$. Similarly, reduction in r_i (decreased growth rate or increased cost) affects equilibrium through $\frac{dx^*}{dr_i}$.

These comparative statics are identical in economic and ecological interpretations, providing testable predictions.

Coevolutionary extension: Parameters r_i themselves may evolve. In ecology, selection on growth rates occurs on longer timescales. In economics, firms invest in cost reduction, changing $r_i \propto 1/c_i$ over time. This suggests a two-timescale model:

$$\text{Fast dynamics: } \frac{dx_i}{dt} = x_i f_i(x; r)$$

$$\text{Slow dynamics: } \frac{dr_i}{d\tau} = g_i(x^*(r), r_i)$$

where τ is the slow timescale and $x^*(r)$ is the fast equilibrium given r . This adaptive dynamics framework unifies evolutionary ecology and endogenous technological change in economics.

Generalized replicator dynamics: In economic contexts, we can derive $\dot{q}_i = q_i(\pi_i - \bar{\pi})$ where $\bar{\pi} = \frac{\sum_j q_j \pi_j}{\sum_j q_j}$ is the market-weighted average profit. This is formally a Lotka-Volterra system with endogenous α_{ij} determined by how firm j 's output affects firm i 's profitability. The interaction coefficients emerge from the demand structure rather than being imposed exogenously, providing microfoundations for the competition matrix.

9. Discussion

We have demonstrated that market equilibria and ecological equilibria are mathematically isomorphic, extending from monopoly (single species) through oligopoly (multi-species) to perfect competition (competitive exclusion). Three key results emerge:

Mathematical equivalence: Static Nash equilibrium conditions transform into dynamical Lotka-Volterra equations under appropriate parameter identification. The critical requirement is linearity (linear demand in economics, linear competitive effects in ecology) which ensures proportional feedback structure. This equivalence is not merely analogical but reflects identical mathematical objects viewed through different interpretive lenses.

Interpretative consequences: The distinction between conscious optimization and evolutionary selection is less fundamental than commonly assumed. Both are gradient ascent processes on fitness/profit landscapes converging to fixed points with identical stability properties. Firms and species face the same formal problem: allocate effort to exploit a limited resource while competing with others doing likewise. The mathematical solution is unique regardless of cognitive apparatus.

Predictive opportunities: The unified framework enables testable predictions across domains. Market concentration patterns should obey ecological stability conditions. Comparative statics under resource shocks (changing K) or efficiency shocks (changing r_i) follow identical trajectories. Industry dynamics under technological change (evolving r_i) should parallel adaptive dynamics in evolving ecosystems. Empirical validation would strengthen the theoretical unification and provide cross-domain forecasting tools.

Future extensions include: (1) Spatial competition via reaction-diffusion equations: $\frac{\partial x_i}{\partial t} = x_i f_i(\mathbf{x}) + D_i \nabla^2 x_i$, modeling geographic markets or habitat fragmentation with diffusion coefficient D_i representing mobility/transportation costs; (2) Stochastic environments through multiplicative noise: $dx_i = x_i f_i(\mathbf{x}) dt + \sigma x_i dW_i$, capturing demand volatility or environmental fluctuations where σ measures uncertainty and dW_i are Wiener processes; (3) Coevolutionary dynamics linking slow parameter evolution $\frac{dr_i}{dt} = g_i(\mathbf{x}^*, r_i)$ to endogenous innovation models (Romer, Aghion-Howitt) where R&D investment changes efficiency over time, with τ as the innovation timescale distinct from market adjustment.

The essential insight remains: competition for limited resources generates universal patterns transcending specific mechanisms. Whether shaped by profit-seeking or natural selection, competitive systems obey the same mathematical laws.

10. Conclusion and Acknowledgments

This work demonstrates that economic markets and ecological systems are governed by a common mathematical law of resource-constrained competition. By establishing a formal isomorphism between Nash equilibria in Cournot markets and steady states in Lotka–Volterra dynamics, we show that optimization and selection are not alternative mechanisms but complementary expressions of the same variational structure. Both processes represent gradient-driven adjustments converging to equilibria where marginal gains vanish under scarcity.

The unified framework developed here reveals that monopoly, oligopoly, and perfect competition correspond precisely to single-species, multi-species, and competitive exclusion regimes in ecology. Economic concentration, ecological coexistence, and technological evolution thus emerge as parallel manifestations of a single organizing geometry that transforms scarcity into structure. The stability conditions that govern coexistence in ecosystems equally determine which market configurations are sustainable, suggesting a deep correspondence between efficiency, diversity, and resilience.

This synthesis provides both a theoretical bridge and a research program. Future work should test empirically whether real markets obey coexistence conditions analogous to those observed in biological communities, and explore whether innovation dynamics replicate adaptive evolution in ecological systems. Ultimately, the proposed law of resource-constrained competition unites growth, adaptation, and equilibrium within a common mathematical architecture, a framework through which the dynamics of economies and ecosystems may be understood as expressions of the same universal principle.

Appendix A: Derivation of Cournot Reaction Functions

Consider the Cournot duopoly with linear inverse demand $P(Q) = a - b(q_1 + q_2)$ where $a > 0$ is the demand intercept and $b > 0$ is the slope parameter. Firms have constant marginal costs $c_1, c_2 \geq 0$ and zero fixed costs. Firm 1's profit is:

$$\pi_1(q_1, q_2) = q_1[a - b(q_1 + q_2) - c_1] = q_1(a - c_1) - bq_1^2 - bq_1q_2$$

The first-order condition for profit maximization:

$$\frac{\partial \pi_1}{\partial q_1} = (a - c_1) - 2bq_1 - bq_2 = 0$$

Solving for q_1 yields firm 1's reaction function:

$$R_1(q_2) = \frac{a - c_1 - bq_2}{2b} = \frac{a - c_1}{2b} - \frac{q_2}{2}$$

By symmetry, firm 2's reaction function is:

$$R_2(q_1) = \frac{a - c_2}{2b} - \frac{q_1}{2}$$

Nash equilibrium (q_1^*, q_2^*) satisfies both reaction functions simultaneously. Substituting $R_2(q_1^*)$ into $R_1(q_2^*)$:

$$q_1^* = \frac{a - c_1}{2b} - \frac{1}{2} \left(\frac{a - c_2}{2b} - \frac{q_1^*}{2} \right)$$

Solving: $q_1^* = \frac{a - 2c_1 + c_2}{3b}$ and by symmetry $q_2^* = \frac{a - 2c_2 + c_1}{3b}$.

For the symmetric case $c_1 = c_2 = c$, equilibrium outputs are $q_1^* = q_2^* = \frac{a - c}{3b}$, total output is $Q^* = \frac{2(a - c)}{3b}$, and equilibrium price is $P^* = \frac{a + 2c}{3}$.

The second-order condition $\frac{\partial^2 \pi_1}{\partial q_1^2} = -2b < 0$ confirms this is a maximum. Stability of the Nash equilibrium under best-response dynamics can be verified by showing that the slope of the reaction function satisfies $|R'_i(q_j)| = \frac{1}{2} < 1$, ensuring convergence.

Appendix B: Stability Analysis of Lotka-Volterra Systems

Consider the two-species Lotka-Volterra competition system:

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1} \right), \quad \frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2} \right)$$

The interior equilibrium (N_1^*, N_2^*) satisfies:

$$N_1^* + \alpha_{12} N_2^* = K_1, \quad N_2^* + \alpha_{21} N_1^* = K_2$$

Solving this linear system:

$$N_1^* = \frac{K_1 - \alpha_{12} K_2}{1 - \alpha_{12} \alpha_{21}}, \quad N_2^* = \frac{K_2 - \alpha_{21} K_1}{1 - \alpha_{12} \alpha_{21}}$$

For coexistence, we require $N_1^*, N_2^* > 0$, which holds when:

$$\frac{K_1}{K_2} > \alpha_{12} \quad \text{and} \quad \frac{K_2}{K_1} > \alpha_{21}$$

These conditions state that each species' carrying capacity must exceed the competitive effect from the other species.

To assess stability, compute the Jacobian at (N_1^*, N_2^*) :

$$J = \begin{pmatrix} -\frac{r_1 N_1^*}{K_1} & -\frac{r_1 \alpha_{12} N_1^*}{K_1} \\ -\frac{r_2 \alpha_{21} N_2^*}{K_2} & -\frac{r_2 N_2^*}{K_2} \end{pmatrix}$$

The trace is $\text{tr}(J) = -\frac{r_1 N_1^*}{K_1} - \frac{r_2 N_2^*}{K_2} < 0$ (always negative). The determinant is:

$$\det(J) = \frac{r_1 r_2 N_1^* N_2^*}{K_1 K_2} (1 - \alpha_{12} \alpha_{21})$$

Stability requires $\det(J) > 0$, which holds when $\alpha_{12} \alpha_{21} < 1$, intraspecific competition must dominate interspecific competition. Combined with coexistence conditions, stable equilibrium requires:

$$\alpha_{12}, \alpha_{21} < 1 \quad \text{and} \quad \frac{K_1}{K_2} > \alpha_{12}, \quad \frac{K_2}{K_1} > \alpha_{21}$$

When $\alpha_{12} \alpha_{21} > 1$, the system exhibits competitive exclusion: whichever species has the advantage (larger K_i/α_{ji} ratio) drives the other extinct.

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